Software Engineering

Tuesday
28 July, 2026

Documentation
operating procedure
No. of programs
Prgrm / Sftwr
set of instruc. to carry-out certain tasks

Phases

Waterfall Model
Req. Analysis
R.A.S. Specifications

- Requirement Gathering
- Design
- Implementation & unit testing
- Integration & system testing
- O & M (operation, maintenance)

Build & fix Model

Applicable for small size project (for generally upto 100 lines of code)

Tuesday
5 Aug, 2026

SDLC Model

software development life cycle

- SRS serves as base (software req. specification)
- { Something to write here }

V- Model

Gathering & understanding customer requirements & preparing
Verification
To ensure customer req. are done or not
Validation
Dynamic tools

- Req. Analysis ↔ Acceptance test design ↔ Acceptance testing
- Static performance
- System Design → Creates high level & detailed sys. design ↔ Sys. test design ↔ System Testing → whole sys as a single unit test
- Architecture Design → you can define overall arch. & preparing integration test design ↔ Integration testing
- Module Design → Design independent modules (Low-level design) ↔ Unit test design ↔ Unit Testing → Individual unit modules testing
- Coding → Develop s/w based on design & follow coding standards & rules

Adv.

- Early detection of defects
- Mapped testing
- Better colab.
- Higher quality product ↑ (reliable & better) can be implemented

Disadv.

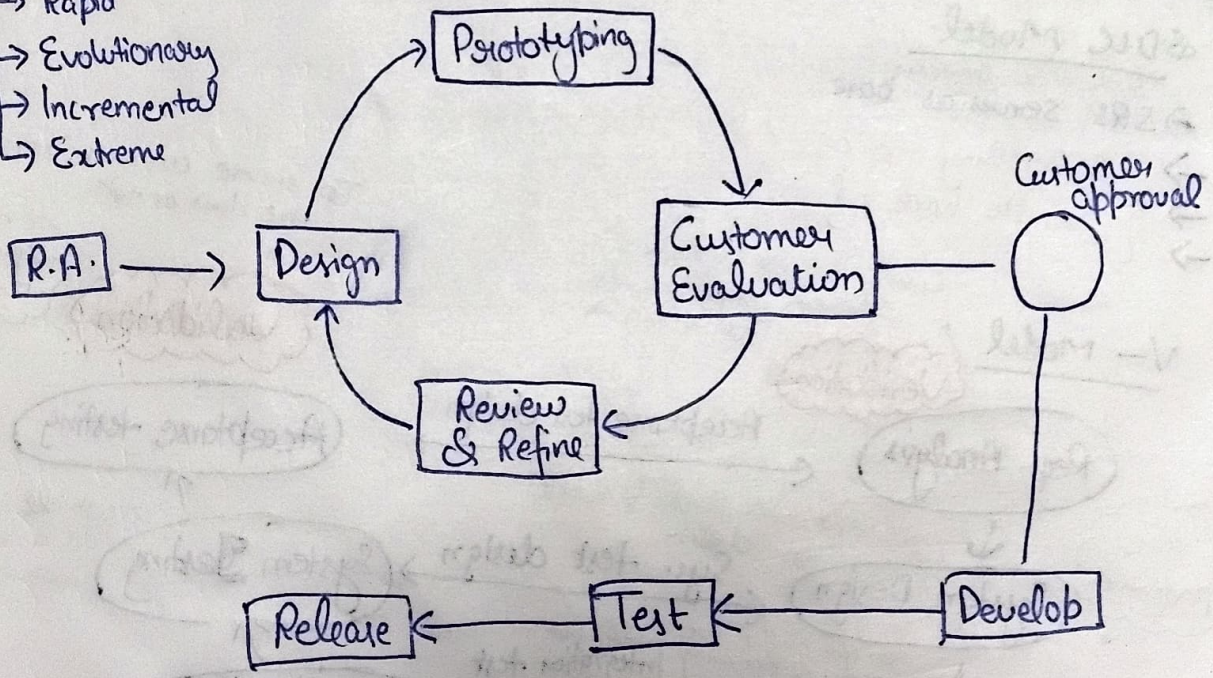
- Difficult accomodate changing req. of customer
- Time - consuming
- Not suitable for complex/high risk projects
- Iterative development not supported

Applications

- Healthcare sys.
- Banking & financial s/w
- Aerospace & aviation
  - ↳ Flight ctrl mngmnt sys.
- Automotive industry

Prototyping Model

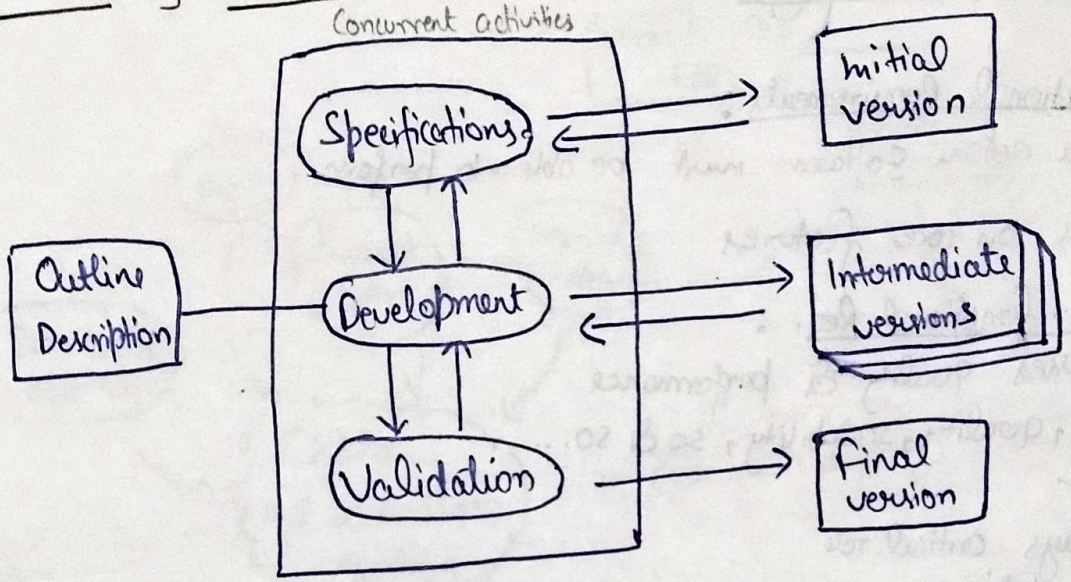
- → Rapid
- → Evolutionary
- → Incremental
- ↳ Extreme



1.) Req. gathering & analysis
2.) Quick design
3.) Build a prototype
4.) Initial user evaluation
5.) Refine prototype
6.) Implement product & maintain

Disadv.
- Time consuming
- Discarded prototype is wastage
- Poor documentation {Something similar to this}

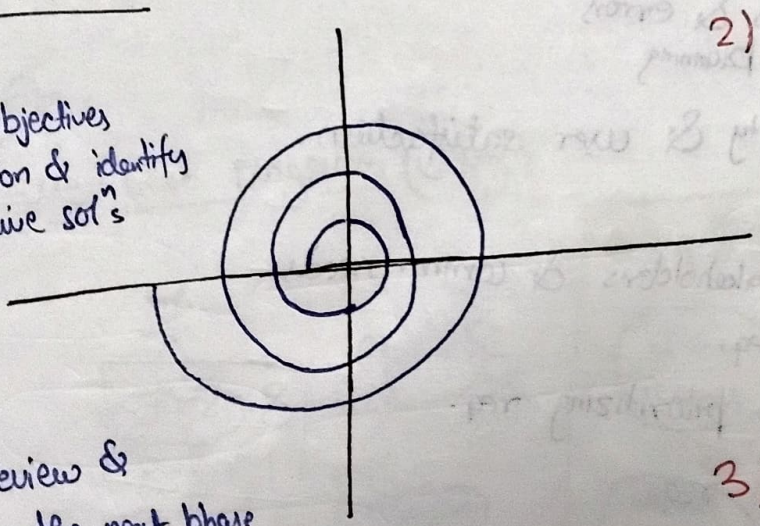
Evolutionary Process Model



Adv
- Risk reduction
- Customer collaboration

Disadv
- Not suitable for small size projects

Spiral Model



1) Objectives determination & identify alternative sol^n^s

2) Identify & resolve risks
Only model to resolve risks

3) Develop next version of product

4) Review & plan for the next phase

Adv
- Risk identified
- Continuous risk evaluation
- More flexible